

Blancmange Analysis

Dyadic Fold Geometry, Tent Expansions, and Fold-Relative Learning

Soumadeep Ghosh

Kolkata, India

Abstract

This paper formulates a possible alternative analysis called *Blancmange Analysis*. Classical real analysis treats the real line as locally straight and studies functions by linear infinitesimals. Blancmange Analysis instead treats the line as recursively folded by dyadic tent maps. Its primitive functions are the dyadic tents

$$\psi(2^n x - k), \quad \psi(x) = \text{dist}(x, \mathbb{Z}),$$

and its central object is the blancmange, or Takagi, function

$$B(x) = \sum_{n=0}^{\infty} 2^{-n} \psi(2^n x).$$

The framework developed here describes real numbers as branch addresses through nested folds, defines fold-relative regularity and differentiation, introduces tent expansions, proves basic convergence and reconstruction results, and gives algorithms for numerical approximation, statistical estimation, and machine-learning models using blancmange features.

The paper ends with “The End”

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1 Introduction

The classical real line is usually analyzed through local linearity. A function $f : \mathbb{R} \rightarrow \mathbb{R}$ is examined by the quotient

$$\frac{f(x+h) - f(x)}{h},$$

and differentiability means that this quotient stabilizes as $h \rightarrow 0$. The blancmange function challenges this picture. It is continuous everywhere but nowhere differentiable, and yet it is built from an extremely simple dyadic mechanism: repeated triangular folding.

Let

$$\psi(x) = \text{dist}(x, \mathbb{Z}).$$

The standard blancmange function is

$$B(x) = \sum_{n=0}^{\infty} 2^{-n} \psi(2^n x).$$

Each term $\psi(2^n x)$ folds the real line into 2^n triangular pieces. Hence B is not merely a pathological example; it encodes a complete dyadic multiscale geometry.

The central proposal of this paper is:

Blancmange Analysis is analysis on dyadic fold geometry.

In this analysis, real numbers are not first described as points of a smooth continuum. They are described as infinite addresses through nested left-right folds. Smoothness is replaced by stability under folding, and linear approximation is replaced by approximation through tent functions.

2 The Dyadic Tent System

Definition 1 (Tent function). *Define*

$$\psi(x) = \text{dist}(x, \mathbb{Z}).$$

Equivalently, for $x \in [0, 1]$,

$$\psi(x) = \begin{cases} x, & 0 \leq x \leq \frac{1}{2}, \\ 1 - x, & \frac{1}{2} \leq x \leq 1, \end{cases}$$

extended periodically with period 1.

Definition 2 (Dyadic tent atoms). *For $n \geq 0$ and $0 \leq k < 2^n$, define the localized dyadic tent*

$$\psi_{n,k}(x) = \psi(2^n x - k).$$

The global dyadic tent at scale n is

$$\psi_n(x) = \psi(2^n x).$$

Definition 3 (Takagi–Landsberg family). *For $0 < \lambda < 1$, define*

$$B_\lambda(x) = \sum_{n=0}^{\infty} \lambda^n \psi(2^n x).$$

The classical blancmange function is the case

$$B(x) = B_{1/2}(x).$$

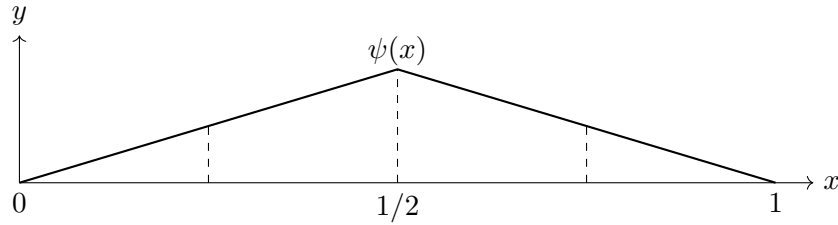


Figure 1: The primitive fold: the tent function $\psi(x) = \text{dist}(x, \mathbb{Z})$ on $[0, 1]$.

3 Real Numbers as Blancmange Addresses

The first reversal is to describe real numbers in terms of the fold system rather than describing the fold system in terms of real numbers.

Let

$$\Sigma = \{0, 1\}^{\mathbb{N}}$$

be the space of infinite binary sequences.

Definition 4 (Dyadic address map). *Define*

$$\pi : \Sigma \rightarrow [0, 1], \quad \pi(\varepsilon_1, \varepsilon_2, \dots) = \sum_{n=1}^{\infty} \frac{\varepsilon_n}{2^n}.$$

The sequence $(\varepsilon_n)_{n \geq 1}$ is called a dyadic fold address.

For non-dyadic x , the address is unique. For dyadic rationals, two addresses exist:

$$0.1000 \dots_2 = 0.0111 \dots_2.$$

Thus

$$[0, 1] \cong \Sigma / \sim,$$

where $\sim$ identifies the two binary expansions of dyadic rationals.

For the whole real line,

$$\mathbb{R} \cong \mathbb{Z} \times \Sigma / \sim.$$

Definition 5 (Fold itinerary). *For $x \in [0, 1]$ not dyadic, define*

$$\varepsilon_{n+1}(x) = \begin{cases} 0, & \{2^n x\} < \frac{1}{2}, \\ 1, & \{2^n x\} > \frac{1}{2}. \end{cases}$$

This records whether x lies on the rising or falling side of the n -th tent wave.

Proposition 1 (Reconstruction from fold itinerary). *For every non-dyadic $x \in [0, 1]$,*

$$x = \sum_{n=1}^{\infty} \frac{\varepsilon_n(x)}{2^n}.$$

For dyadic x , the same formula holds after quotienting the two binary representatives.

Proof. The fold itinerary is exactly the binary expansion of x . At scale n , the condition

$$\{2^{n-1}x\} < \frac{1}{2}$$

places x in the left half of a dyadic interval of length $2^{-(n-1)}$, while the opposite condition places it in the right half. Thus each ε_n selects a dyadic child interval. The nested intervals have lengths 2^{-n} , hence their intersection contains exactly x . Therefore x is reconstructed by its binary expansion. $\square$

4 Axioms of Blancmange Analysis

Axiom 1 (Fold primacy). *Every local phenomenon is observed through dyadic folding. The primitive local probes are the functions*

$$\psi(2^n x - k).$$

Axiom 2 (Scale descent). *The limiting process $h \rightarrow 0$ is replaced by descent through dyadic scales:*

$$h_n = 2^{-n} \rightarrow 0.$$

Axiom 3 (Fold-relative comparison). *The variation of a function f is measured relative to the variation of a fold function B_λ , not only relative to Euclidean displacement.*

Classical Analysis	Blancmange Analysis	Interpretation
Point x	Address (ε_n)	Branch sequence
Increment $h \rightarrow 0$	Scale $2^{-n} \rightarrow 0$	Dyadic descent
Line segment	Tent segment	Folded local geometry
Derivative df/dx	Fold derivative $D_B f$	Slope against B
Smoothness	Fold regularity	Stability under folding
Taylor expansion	Tent expansion	Dyadic fold approximation
Fourier modes	Tent atoms	Multiscale folded basis

Table 1: Dictionary between classical analysis and Blancmange Analysis.

5 Basic Analytic Properties

Theorem 1 (Uniform convergence). *For every $0 < \lambda < 1$, the series*

$$B_\lambda(x) = \sum_{n=0}^{\infty} \lambda^n \psi(2^n x)$$

converges uniformly on $\mathbb{R}$. Consequently B_λ is continuous.

Proof. Since

$$0 \leq \psi(x) \leq \frac{1}{2}$$

for all x , we have

$$|\lambda^n \psi(2^n x)| \leq \frac{1}{2} \lambda^n.$$

Because

$$\sum_{n=0}^{\infty} \frac{1}{2} \lambda^n = \frac{1}{2(1-\lambda)} < \infty,$$

the Weierstrass M -test gives uniform convergence. Each summand is continuous, so the uniform limit is continuous. $\square$

Definition 6 (Blancmange scale quotient). *Let $f : [0, 1] \rightarrow \mathbb{R}$. For an admissible point x , define the n -th blancmange quotient by*

$$Q_n^B f(x) = \frac{f(x + 2^{-n}) - f(x)}{B(x + 2^{-n}) - B(x)},$$

whenever the denominator is nonzero.

Definition 7 (Blancmange derivative). *If the limit exists along all sufficiently large admissible n , define*

$$D_B f(x) = \lim_{n \rightarrow \infty} Q_n^B f(x).$$

More generally,

$$D_{B_\lambda} f(x) = \lim_{n \rightarrow \infty} \frac{f(x + 2^{-n}) - f(x)}{B_\lambda(x + 2^{-n}) - B_\lambda(x)}.$$

Remark 1. *The definition is scale-directed rather than direction-directed. Classical derivatives test all small h . Blancmange derivatives test the canonical dyadic scales of the fold geometry.*

Proposition 2 (Chain rule for fold-observable functions). *Let $f(x) = F(B(x))$, where F is differentiable at $B(x)$. If $D_B f(x)$ exists, then*

$$D_B f(x) = F'(B(x)).$$

Proof. For admissible n ,

$$Q_n^B f(x) = \frac{F(B(x + 2^{-n})) - F(B(x))}{B(x + 2^{-n}) - B(x)}.$$

As $n \rightarrow \infty$, continuity of B gives

$$B(x + 2^{-n}) \rightarrow B(x).$$

Therefore the quotient converges to $F'(B(x))$. $\square$

6 Fold-Regularity

Definition 8 (Fold metric). *For $\eta > 0$, define*

$$d_B(x, y) = |x - y| + \eta|B(x) - B(y)|.$$

The ordinary term $|x - y|$ prevents degeneracy, while the second term introduces fold geometry.

Definition 9 (Blancmange-Hölder regularity). *A function $f : [0, 1] \rightarrow \mathbb{R}$ is B -Hölder of order $\alpha \in (0, 1]$ if there exists $C > 0$ such that*

$$|f(x) - f(y)| \leq C d_B(x, y)^\alpha$$

for all $x, y \in [0, 1]$.

Proposition 3. *If f is ordinary Hölder of order α , then f is B -Hölder of order α .*

Proof. Since

$$|x - y| \leq d_B(x, y),$$

ordinary Hölder continuity gives

$$|f(x) - f(y)| \leq C|x - y|^\alpha \leq C d_B(x, y)^\alpha.$$

□

Remark 2. *The converse need not hold in a strong geometric sense. A function may vary sharply in Euclidean space while remaining stable relative to the folded graph metric.*

7 Tent Expansions

Definition 10 (Global tent expansion). *A global blancmange expansion of f is a representation of the form*

$$f(x) \sim a_0 + \sum_{n=0}^{\infty} a_n \psi(2^n x).$$

Definition 11 (Localized tent expansion). *A localized tent expansion is*

$$f(x) \sim a_0 + \sum_{n=0}^{\infty} \sum_{k=0}^{2^n-1} a_{n,k} \psi(2^n x - k).$$

Let

$$\mathcal{B} = \text{span}\{\psi(2^n x - k) : n \geq 0, 0 \leq k < 2^n\}.$$

Theorem 2 (Density intuition). *The space $\mathcal{B}$ contains the classical continuous piecewise-linear dyadic functions on $[0, 1]$. Consequently its uniform closure contains all continuous functions on $[0, 1]$.*

Proof. Continuous piecewise-linear functions on dyadic partitions can be written using shifted tent, or hat, functions. The atoms $\psi(2^n x - k)$ generate such dyadic hat functions after restriction and finite linear combination. Continuous piecewise-linear dyadic functions are uniformly dense in $C[0, 1]$ by uniform continuity and linear interpolation on sufficiently fine partitions. Hence the closure of $\mathcal{B}$ contains $C[0, 1]$. □

8 Algorithms

Algorithm 1 (Numerical evaluation of B_λ). *Given $x \in [0, 1]$, $0 < \lambda < 1$, and truncation level N :*

1. $S \leftarrow 0$.
2. *For $n = 0, 1, \dots, N$:*
 - $u \leftarrow 2^n x - \lfloor 2^n x \rfloor$.
 - $t \leftarrow \min(u, 1 - u)$.
 - $S \leftarrow S + \lambda^n t$.
3. *Return S .*

The truncation error satisfies

$$|B_\lambda(x) - S| \leq \sum_{n=N+1}^{\infty} \frac{1}{2} \lambda^n = \frac{\lambda^{N+1}}{2(1-\lambda)}.$$

Algorithm 2 (Fold-address reconstruction). *Given bits $(\varepsilon_1, \dots, \varepsilon_N)$, approximate the real number by*

$$x_N = \sum_{n=1}^N \frac{\varepsilon_n}{2^n}.$$

The error is bounded by

$$|x - x_N| \leq 2^{-N}.$$

Algorithm 3 (Least-squares tent regression). *Given observations $(x_i, y_i)_{i=1}^m$, choose a maximum scale N and solve*

$$\min_{\beta} \sum_{i=1}^m \left(y_i - \beta_0 - \sum_{n=0}^N \sum_{k=0}^{2^n-1} \beta_{n,k} \psi(2^n x_i - k) \right)^2.$$

The fitted model is

$$\hat{f}_N(x) = \hat{\beta}_0 + \sum_{n=0}^N \sum_{k=0}^{2^n-1} \hat{\beta}_{n,k} \psi(2^n x - k).$$

9 Statistics and Data Science

Suppose data are generated by

$$Y_i = f(X_i) + \varepsilon_i, \quad \mathbb{E}[\varepsilon_i | X_i] = 0, \quad \text{Var}(\varepsilon_i | X_i) = \sigma^2.$$

In Blancmange Analysis, a natural feature map is

$$\Phi_N(x) = (1, \psi(x), \psi(2x), \dots, \psi(2^N x)).$$

A richer localized map is

$$\Phi_N^{\text{loc}}(x) = (1, \psi(2^n x - k))_{0 \leq n \leq N, 0 \leq k < 2^n}.$$

Definition 12 (Blancmange covariance kernel). *Define*

$$K_N(x, y) = \sum_{n=0}^N \gamma_n \psi(2^n x) \psi(2^n y), \quad \gamma_n \geq 0.$$

The infinite kernel is

$$K(x, y) = \sum_{n=0}^{\infty} \gamma_n \psi(2^n x) \psi(2^n y),$$

whenever the series converges.

Proposition 4. *If $\gamma_n \geq 0$ and $\sum_n \gamma_n < \infty$, then K is a positive semidefinite kernel.*

Proof. For any $x_1, \dots, x_m$ and $c_1, \dots, c_m$,

$$\sum_{i,j=1}^m c_i c_j K(x_i, x_j) = \sum_{n=0}^{\infty} \gamma_n \left(\sum_{i=1}^m c_i \psi(2^n x_i) \right)^2 \geq 0.$$

Absolute convergence follows from $\psi \leq 1/2$ and $\sum_n \gamma_n < \infty$. □

Task	Classical Tool	Blancmange Tool
Regression	Polynomials, splines	Tent regression
Classification	Kernel machines	Fold kernels
Denoising	Wavelets, filters	Dyadic tent shrinkage
Feature engineering	Fourier features	Fold-address features
Anomaly detection	Residual thresholds	Fold-scale residuals
Time series	ARMA, spectra	Fold-scale decomposition

Table 2: Statistical and machine-learning uses of blancmange features.

10 AI and Machine Learning

A neural model can incorporate fold geometry by using blancmange features as an input layer. For $x \in [0, 1]$, define

$$z(x) = (x, B_\lambda(x), \psi(x), \psi(2x), \dots, \psi(2^N x)).$$

A feedforward model becomes

$$\hat{y} = F_\theta(z(x)).$$

One may also define a fold-regularized objective:

$$\mathcal{L}(\theta) = \frac{1}{m} \sum_{i=1}^m \ell(F_\theta(z(x_i)), y_i) + \rho \sum_{i,j} \frac{|F_\theta(z(x_i)) - F_\theta(z(x_j))|^2}{d_B(x_i, x_j)^2 + \delta},$$

where $\rho > 0$ is a regularization strength and $\delta > 0$ prevents division by zero.

This encourages the learned function to be stable not merely under Euclidean closeness but under closeness in the folded blancmange geometry.

11 A Diagram of Fold Descent

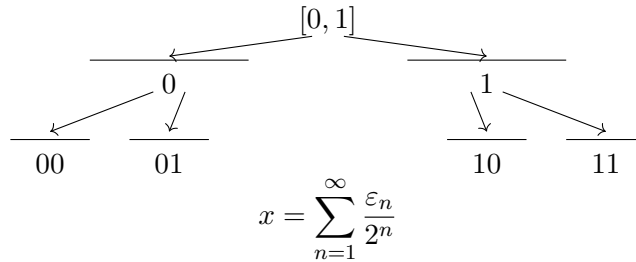


Figure 2: A real number as an infinite branch address through dyadic fold descent.

12 Interpretation

Blancmange Analysis is not a rejection of real analysis. It is a change of primitive geometry. Classical analysis begins from straightness and treats roughness as exceptional. Blancmange Analysis begins from recursive folding and treats smoothness as a special low-complexity case.

Its guiding slogan is:

Do not analyze functions only by straightening curves; analyze them by folding the line.

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Glossary

Blancmange function The continuous nowhere-differentiable function

$$B(x) = \sum_{n=0}^{\infty} 2^{-n} \psi(2^n x).$$

Tent function The function

$$\psi(x) = \text{dist}(x, \mathbb{Z}),$$

which forms triangular waves over unit intervals.

Dyadic fold The operation of observing x through $\psi(2^n x)$, producing 2^n triangular folds.

Fold address The binary sequence (ε_n) determining the nested dyadic location of x .

Blancmange derivative A derivative-like object comparing increments of f to increments of B , rather than to increments of x .

Fold regularity A notion of regularity measured relative to the graph geometry of B or B_λ .

Tent expansion A function expansion in terms of atoms $\psi(2^n x - k)$.

Fold kernel A positive semidefinite kernel built from dyadic tent features.

Fold-relative learning Machine learning in which features, losses, or regularizers are defined using blancmange geometry.

The End